

Cody Pych

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

(52)

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Quiz #4

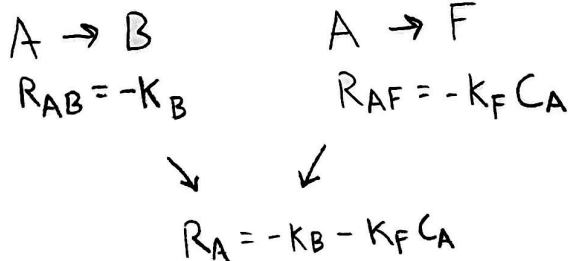
November 18, 2025

Open Notes and Book

Duration: 40 minutes

Consider the irreversible conversion of species A into B and F through the reactions $A \rightarrow B$ and $A \rightarrow F$ in a stationary liquid film of thickness L . Determine C_A throughout the film, if the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentration of A at $y = 0$ is specified as C_0 . The homogeneous reactions follow zeroth and first order kinetics with rates of reaction $R_{AB} = -k_B$ and $R_{AF} = -k_F C_A$, respectively. The diffusivity of the species A is D_A .

Find the overall rate of reactant consumption.



$$C_A|_{y=0} = C_0$$

$$C_A|_{y=L} = 0$$

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$$D_A \frac{d^2 C_A}{dy^2} - K_F C_A - K_B = 0$$

$$D_A \frac{d^2 C_A}{dy^2} - K_F \left(C_A + \frac{K_B}{K_F} \right) = 0$$

$$D_A \frac{d^2}{dy^2} \left(C_A + \frac{K_B}{K_F} \right) - K_F \left(C_A + \frac{K_B}{K_F} \right) = 0$$

$$C_A(y) = H e^{\sqrt{\frac{K_F}{D_A}} y} + J e^{-\sqrt{\frac{K_F}{D_A}} y} - \frac{K_B}{K_F}$$

$$C_A(y=0) = H + J = C_0$$

$$C_A(y=L) = H e^{\sqrt{\frac{K_F}{D_A}} L} + J e^{-\sqrt{\frac{K_F}{D_A}} L}$$